🌐 MODULE 7 — Hybrid Education Model Generator for India

ML-Based Comparative Analysis of Global Education Systems

> **Goal:** Using ML predictions and comparative analysis, recommend an optimized **Hybrid Education Reform Model** for India by selectively combining the best features from Finland, Canada, Singapore, and Germany.

Why a Hybrid model?
India should NOT blindly copy one country. Each country excels in different dimensions:

- FI **Finland** → Teacher quality, student well-being, low failures
- CA **Canada** → Government investment, school quality
- SG **Singapore** → Academic rigor, school infrastructure
- DE **Germany** → Teacher development, structured learning

Steps in this notebook:

1. Install & import libraries
2. Upload dataset
3. Train ML model & extract country profiles
4. Analyze India's current gaps
5. Build the Hybrid model (blend best features)
6. Predict performance improvement
7. Phased reform roadmap simulation
8. Visualizations (5 charts)
9. Policy recommendations
10. Download all results

⚙️ Step 1 — Install & Import Libraries

```
!pip install -q scikit-learn pandas numpy matplotlib seaborn
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
```

```
import matplotlib.gridspec as gridspec
from matplotlib.patches import FancyArrowPatch
import seaborn as sns
from sklearn.ensemble import RandomForestRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
import warnings
warnings.filterwarnings('ignore')
np.random.seed(42)
print('✅ Libraries imported!')
```

```
✅ Libraries imported!
```

📂 Step 2 — Upload Dataset

```
from google.colab import files
import io

print('📂 Upload your student_performance_global.csv...')
uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(io.BytesIO(uploaded[filename]))

print(f"\n✅ Loaded: {df.shape[0]} rows x {df.shape[1]} columns")
print(f"Countries: {sorted(df['country'].unique().tolist())}")
df.head(3)
```

```
📂 Upload your student_performance_global.csv...
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving student_performance_global.csv to student_performance_global.csv

✅ Loaded: 650 rows x 41 columns
Countries: ['Canada', 'China', 'Finland', 'Germany', 'India', 'Japan', 'Singa
```

	student_id	country	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjc
0	1	India	F	15	R	LE3	T	3.90	2.80	oth
1	2	India	M	19	U	LE3	T	4.09	4.65	teach
2	3	India	M	18	U	LE3	T	3.42	3.77	oth

3 rows × 41 columns

🤖 Step 3 — Train ML Model & Build Country Profiles

```
FEATURES = [
    'math_score_index', 'reading_score_index', 'science_score_index',
    'school_quality_index', 'govt_edu_spend_index', 'teacher_ratio_index',
```

```
    'student_wellbeing_index', 'studytime', 'failures', 'absences',
]
FEATURE_LABELS = {
    'math_score_index':        'Math Score',
    'reading_score_index':     'Reading Score',
    'science_score_index':     'Science Score',
    'school_quality_index':    'School Quality',
    'govt_edu_spend_index':    'Govt. Spend',
    'teacher_ratio_index':     'Teacher Quality',
    'student_wellbeing_index': 'Student Well-being',
    'studytime':               'Study Time',
    'failures':                'Past Failures',
    'absences':                'Absences',
}
NEGATIVE_FEATURES = ['failures', 'absences']
TARGET = 'G3'

# Validate columns
FEATURES = [f for f in FEATURES if f in df.columns]

# Train Random Forest
model_df = df[FEATURES + [TARGET, 'country']].dropna()
X = model_df[FEATURES]; y = model_df[TARGET]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0

rf = RandomForestRegressor(n_estimators=200, random_state=42, max_depth=8)
rf.fit(X_train, y_train)
r2   = r2_score(y_test, rf.predict(X_test))
rmse = np.sqrt(mean_squared_error(y_test, rf.predict(X_test)))
print(f'✅ Random Forest trained — R² = {r2:.4f} | RMSE = {rmse:.4f}')

fi_norm = pd.Series(rf.feature_importances_ / rf.feature_importances_.sum(),

# Country profiles (mean of 65 students each)
country_profiles = df.groupby('country')[FEATURES].mean()
country_profiles['G3_actual'] = df.groupby('country')[TARGET].mean()

# Predict G3 for each country
cp_X = scaler.transform(country_profiles[FEATURES])
country_profiles['G3_predicted'] = rf.predict(cp_X)

print('\n📊 Country Profiles (key features):')
display_cols = ['teacher_ratio_index','student_wellbeing_index',
                'school_quality_index','govt_edu_spend_index',
                'failures','absences','G3_actual','G3_predicted']
display(country_profiles[display_cols].round(3).style
    .background_gradient(subset=['G3_actual','teacher_ratio_index','student_
    .background_gradient(subset=['failures','absences'], cmap='YlOrRd')
    .highlight_max(subset=['G3_actual'], color='#DCFCE7')
    .highlight_min(subset=['G3_actual'], color='#FEE2E2')
)
```

```
✅ Random Forest trained — R² = 0.9046 | RMSE = 0.4359

📊 Country Profiles (key features):
```

country	teacher_ratio_index	student_wellbeing_index	school_quality_index
Canada	8.836000	8.057000	8.989000
China	6.502000	4.633000	7.922000
Finland	9.002000	8.860000	9.543000
Germany	8.569000	7.423000	8.394000
India	3.455000	4.966000	4.864000
Japan	8.049000	4.995000	9.472000
Singapore	8.580000	5.467000	9.704000
South Korea	7.045000	3.953000	8.940000
UK	8.038000	7.088000	8.488000

🔍 Step 4 — India Gap Analysis

For every feature, we measure how far India is from the **best-performing country** in that dimension.

```
india    = country_profiles.loc['India', FEATURES]
finland  = country_profiles.loc['Finland', FEATURES]
canada   = country_profiles.loc['Canada', FEATURES]
singapore= country_profiles.loc['Singapore', FEATURES]
germany  = country_profiles.loc['Germany', FEATURES]

# For each feature: who is best?
gap_rows = []
for f in FEATURES:
    if f in NEGATIVE_FEATURES:
        best_country = country_profiles[f].idxmin()
        best_val     = country_profiles[f].min()
        gap          = india[f] - best_val   # positive = India is worse
    else:
        best_country = country_profiles[f].idxmax()
        best_val     = country_profiles[f].max()
        gap          = best_val - india[f]   # positive = India is worse

    gap_rows.append({
        'Feature':         FEATURE_LABELS[f],
        'India Value':     round(india[f], 3),
        'Best Country':    best_country,
        'Best Value':      round(best_val, 3),
```

```
        'Gap (India lag)': round(gap, 3),
        'FI Weight':       round(fi_norm[f], 4),
        'Weighted Gap':    round(gap * fi_norm[f], 4),
        'Impact':          'NEGATIVE factor' if f in NEGATIVE_FEATURES else
    })

gap_df = pd.DataFrame(gap_rows).sort_values('Weighted Gap', ascending=False)
gap_df.index = range(1, len(gap_df)+1)

print('📋 India Feature Gap Analysis (sorted by Weighted Gap):\n')
display(gap_df.style
    .background_gradient(subset=['Gap (India lag)'], cmap='YlOrRd')
    .background_gradient(subset=['Weighted Gap'], cmap='Reds')
    .format({'India Value':'{:.3f}','Best Value':'{:.3f}',
             'Gap (India lag)':'{:.3f}','FI Weight':'{:.4f}','Weighted Gap':
)

print('\n💡 Top 3 priority gaps for India:')
for i, row in gap_df.head(3).iterrows():
    print(f'   {i}. {row["Feature"]} — lag of {row["Gap (India lag)"]:.3f} b
```

```
📋 India Feature Gap Analysis (sorted by Weighted Gap):
```

	Feature	India Value	Best Country	Best Value	Gap (India lag)	FI Weight	Weighted Gap	Impact
1	Science Score	4.265	Singapore	9.260	4.994	0.4073	2.0344	POSITIVE factor
2	Math Score	4.538	Singapore	9.483	4.946	0.3332	1.6482	POSITIVE factor
3	Past Failures	5.959	Singapore	1.274	4.685	0.0827	0.3872	NEGATIVE factor
4	Reading Score	4.259	Singapore	9.040	4.781	0.0398	0.1904	POSITIVE factor
5	School Quality	4.864	Singapore	9.704	4.840	0.0319	0.1546	POSITIVE factor
6	Govt. Spend	4.088	Finland	8.942	4.854	0.0274	0.1328	POSITIVE factor
7	Teacher	3.455	Finland	9.002	5.547	0.0230	0.1278	POSITIVE factor
8	Student Well-	4.966	Finland	8.860	3.894	0.0257	0.1001	POSITIVE

🔬 Step 5 — Build the Hybrid Education Model

The hybrid model blends India's current values with best-practice values from reference countries. Each feature is sourced from the country that excels in that specific dimension:

Source Country	What India borrows
FI Finland	Teacher quality, student well-being, reducing failures/absences
CA Canada	Government education spending, school quality
SG Singapore	Academic score standards (math, reading, science)
DE Germany	Teacher development structure, school infrastructure

Blend formula:

```
Hybrid value = India value + alpha x (Source country value - India value)
```

where `alpha` = how much India adopts from that country (0 = no change, 1 = fully adopt).

```
# ── Hybrid model configuration ──────────────────────────────────────────
# (feature, source_country, blend_alpha, rationale)
HYBRID_CONFIG = [
    ('teacher_ratio_index',    'Finland',   0.5,  'Adopt Finland teacher tra
    ('student_wellbeing_index','Finland',   0.6,  'Adopt Finland low-stress,
    ('govt_edu_spend_index',   'Canada',    0.4,  'Increase govt spending to
    ('school_quality_index',   'Canada',    0.4,  'Improve school infrastruc
    ('math_score_index',       'Singapore', 0.3,  'Strengthen math curriculu
    ('reading_score_index',    'Singapore', 0.3,  'Strengthen reading standa
    ('science_score_index',    'Singapore', 0.3,  'Strengthen science curric
    ('failures',               'Finland',   0.5,  'Reduce failure rate using
    ('absences',               'Finland',   0.5,  'Reduce absenteeism throug
    ('studytime',              'Germany',   0.3,  'Structured study time lik
]

# Build hybrid profile
hybrid = india.copy()
hybrid_breakdown = []

for feat, source, alpha, rationale in HYBRID_CONFIG:
    source_val  = country_profiles.loc[source, feat]
    india_val   = india[feat]
    hybrid_val  = india_val + alpha * (source_val - india_val)
    hybrid[feat] = hybrid_val
    hybrid_breakdown.append({
        'Feature':       FEATURE_LABELS[feat],
        'India (now)':   round(india_val, 3),
        'Source':        source,
        'Source Value':  round(source_val, 3),
        'Blend Alpha':   alpha,
        'Hybrid Target': round(hybrid_val, 3),
        'Change':        round(hybrid_val - india_val, 3),
        'Rationale':     rationale,
    })

hybrid_df = pd.DataFrame(hybrid_breakdown)
hybrid_df.index = range(1, len(hybrid_df)+1)

print('📋 Hybrid Model — Feature-by-Feature Plan:\n')
```

```
display(hybrid_df.style
    .background_gradient(subset=['Change'], cmap='RdYlGn')
    .format({'India (now)':'{:.3f}','Source Value':'{:.3f}',
             'Blend Alpha':'{:.1f}','Hybrid Target':'{:.3f}','Change':'{:+.3
)

# Predict performance
india_pred  = rf.predict(scaler.transform([india[FEATURES]]))[0]
hybrid_pred = rf.predict(scaler.transform([hybrid[FEATURES]]))[0]
improvement = ((hybrid_pred - india_pred) / india_pred) * 100

print('\n' + '='*55)
print(f'  India current predicted G3  : {india_pred:.4f}')
print(f'  Hybrid model predicted G3   : {hybrid_pred:.4f}')
print(f'  Predicted improvement       : +{improvement:.2f}%')
print('='*55)

# Compare to all countries
print('\n🌍 How hybrid India compares to other countries:')
all_preds = country_profiles['G3_predicted'].copy()
all_preds['India (Hybrid)'] = hybrid_pred
all_preds_sorted = all_preds.sort_values(ascending=False)
for i, (country, val) in enumerate(all_preds_sorted.items(), 1):
    marker = ' ◀ Hybrid India' if country == 'India (Hybrid)' else (' ◀ Inc
    print(f'  {i:2}. {country:<20} {val:.4f}{marker}')
```

```
📋 Hybrid Model — Feature-by-Feature Plan:
```

	Feature	India (now)	Source	Source Value	Blend Alpha	Hybrid Target	Change	Rationale
1	Teacher Quality	3.455	Finland	9.002	0.5	6.229	+2.773	Adopt Finland teacher training standards
2	Student Well-being	4.966	Finland	8.860	0.6	7.302	+2.336	Adopt Finland low-stress, holistic model
3	Govt. Spend	4.088	Canada	8.436	0.4	5.827	+1.739	Increase govt spending toward Canada levels
4	School Quality	4.864	Canada	8.989	0.4	6.514	+1.650	Improve school infrastructure like Canada
5	Math Score	4.538	Singapore	9.483	0.3	6.021	+1.484	Strengthen math curriculum like Singapore
6	Reading Score	4.259	Singapore	9.040	0.3	5.693	+1.434	Strengthen reading standards like Singapore
7	Science Score	4.265	Singapore	9.260	0.3	5.764	+1.498	Strengthen science curriculum like Singapore
8	Past Failures	5.959	Finland	1.512	0.5	3.735	-2.223	Reduce failure rate using formative assessment

📅 Step 6 — Phased Reform Roadmap

Real reform cannot happen overnight. We simulate a **4-phase roadmap** showing India's predicted improvement over time.

```
# Phased reform — progressively move from India to hybrid
PHASES = [
    ('Phase 1 (0-2 yrs)',  0.20, 'Policy foundation: teacher recruitment & w
    ('Phase 2 (2-4 yrs)',  0.40, 'Infrastructure: school quality & spending
    ('Phase 3 (4-7 yrs)',  0.70, 'Curriculum reform: reduce failures, improv
    ('Phase 4 (7-10 yrs)', 1.00, 'Full hybrid model: all reforms consolidate
]
```

```
phase_results = []
baseline_pred = india_pred

print('📅 Phased Reform Simulation:\n')
print(f'{"Phase":<25} {"Alpha":>7} {"Pred G3":>10} {"Improvement":>13}   Des
print('-'*90)

# Baseline row
print(f'{"Baseline (India now)":<25} {0.00:>7.2f} {india_pred:>10.4f} {0.00:
phase_results.append({'Phase':'Baseline','Alpha':0,'Predicted G3':india_pred

for phase_name, alpha, desc in PHASES:
    phase_profile = india.copy()
    for feat, source, base_alpha, _ in HYBRID_CONFIG:
        source_val = country_profiles.loc[source, feat]
        phase_profile[feat] = india[feat] + (alpha * base_alpha) * (source_v
        if feat in NEGATIVE_FEATURES:
            phase_profile[feat] = max(0, phase_profile[feat])

    phase_pred = rf.predict(scaler.transform([phase_profile[FEATURES]]))[0]
    imp        = ((phase_pred - baseline_pred) / baseline_pred) * 100

    print(f'{phase_name:<25} {alpha:>7.2f} {phase_pred:>10.4f} {imp:>+12.2f}
    phase_results.append({'Phase': phase_name, 'Alpha': alpha,
                          'Predicted G3': phase_pred, 'Improvement %': round

phase_df = pd.DataFrame(phase_results)

print('\n💡 By Phase 4 (full reform), India achieves a')
final_imp = phase_df.iloc[-1]['Improvement %']
print(f'   {final_imp:.2f}% improvement in predicted educational performance
```

```
📅 Phased Reform Simulation:

Phase                       Alpha    Pred G3   Improvement   Description
------------------------------------------------------------------------------
Baseline (India now)         0.00     4.7350         0.00%   Current state
Phase 1 (0-2 yrs)            0.20     4.6381        -2.05%   Policy foundatio
Phase 2 (2-4 yrs)            0.40     5.0387        +6.41%   Infrastructure:
Phase 3 (4-7 yrs)            0.70     5.4265       +14.60%   Curriculum refor
Phase 4 (7-10 yrs)           1.00     6.1645       +30.19%   Full hybrid mode

💡 By Phase 4 (full reform), India achieves a
   30.19% improvement in predicted educational performance.
```

📊 Step 7 — Visualizations (5 Charts)

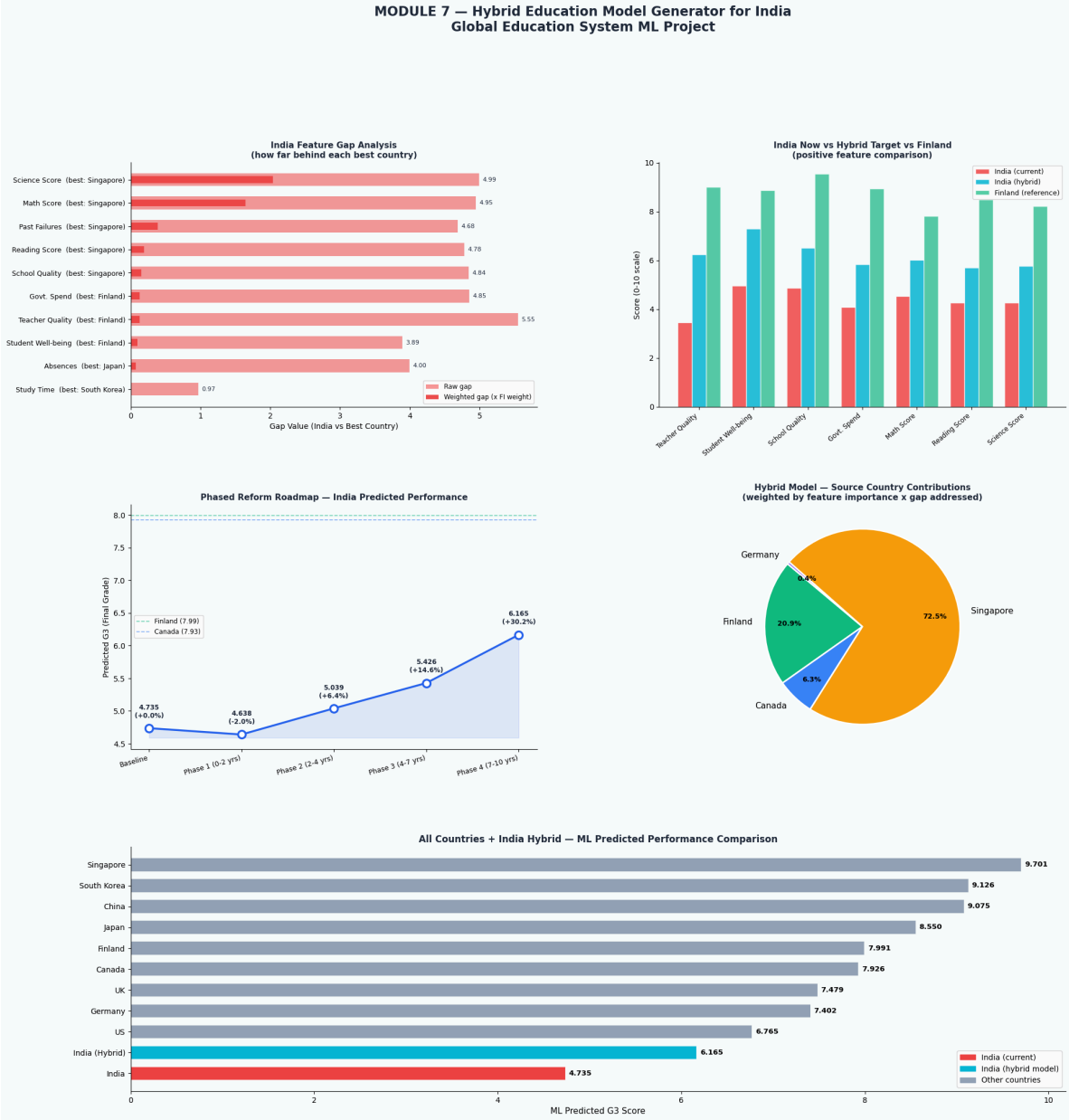
```
SOURCE_COLORS = {
    'Finland':   '#10B981',
    'Canada':    '#3B82F6',
    'Singapore': '#F59E0B',
    'Germany':   '#8B5CF6',
    'India':     '#EF4444',
```

```
    'Hybrid':    '#06B6D4',
}

fig = plt.figure(figsize=(22, 22))
fig.patch.set_facecolor('#F8FAFC')
gs  = gridspec.GridSpec(3, 2, figure=fig, hspace=0.40, wspace=0.30)
fig.suptitle(
    'MODULE 7 — Hybrid Education Model Generator for India\n'
    'Global Education System ML Project',
    fontsize=17, fontweight='bold', color='#1E293B', y=1.01
)

# ── Chart 1: India Gap Analysis bar ──────────────────────────────────
ax1 = fig.add_subplot(gs[0, 0])
ax1.set_facecolor('#F8FAFC')

gap_sorted = gap_df.sort_values('Weighted Gap', ascending=True)
feat_names  = gap_sorted['Feature'].tolist()
gap_vals    = gap_sorted['Gap (India lag)'].tolist()
wgap_vals   = gap_sorted['Weighted Gap'].tolist()
best_country_labels = gap_sorted['Best Country'].tolist()

y_pos = np.arange(len(feat_names))
ax1.barh(y_pos, gap_vals, color='#EF4444', alpha=0.55, height=0.55, label='R
ax1.barh(y_pos, wgap_vals, color='#EF4444', alpha=0.95, height=0.30, label='
ax1.set_yticks(y_pos)
ax1.set_yticklabels(
    [f'{n}  (best: {c})' for n, c in zip(feat_names, best_country_labels)],
    fontsize=9
)
for i, (gv, wg) in enumerate(zip(gap_vals, wgap_vals)):
    ax1.text(gv + 0.05, i, f'{gv:.2f}', va='center', fontsize=8, color='#1E2
ax1.set_xlabel('Gap Value (India vs Best Country)', fontsize=10)
ax1.set_title('India Feature Gap Analysis\n(how far behind each best country
              fontsize=11, fontweight='bold', color='#1E293B')
ax1.legend(fontsize=8.5)
ax1.spines[['top','right']].set_visible(False)

# ── Chart 2: India vs Hybrid radar / grouped bar ─────────────────────
ax2 = fig.add_subplot(gs[0, 1])
ax2.set_facecolor('#F8FAFC')

# Normalize features to 0-10 scale for display (higher = always better)
display_features = [
    'teacher_ratio_index','student_wellbeing_index','school_quality_index',
    'govt_edu_spend_index','math_score_index','reading_score_index',
    'science_score_index'
]
display_labels = [FEATURE_LABELS[f] for f in display_features]

india_vals  = [india[f] for f in display_features]
hybrid_vals = [hybrid[f] for f in display_features]
finland_vals_list = [country_profiles.loc['Finland', f] for f in display_fea

x = np.arange(len(display_features))
```

```
w = 0.26
ax2.bar(x - w, india_vals,      w, label='India (current)', color='#EF4444',
ax2.bar(x,     hybrid_vals,     w, label='India (hybrid)',  color='#06B6D4',
ax2.bar(x + w, finland_vals_list, w, label='Finland (reference)', color='#10

ax2.set_xticks(x)
ax2.set_xticklabels(display_labels, rotation=40, ha='right', fontsize=8.5)
ax2.set_ylabel('Score (0-10 scale)', fontsize=10)
ax2.set_title('India Now vs Hybrid Target vs Finland\n(positive feature comp
              fontsize=11, fontweight='bold', color='#1E293B')
ax2.legend(fontsize=9)
ax2.spines[['top','right']].set_visible(False)

# ── Chart 3: Phased roadmap line chart ──────────────────────────────────
ax3 = fig.add_subplot(gs[1, 0])
ax3.set_facecolor('#F8FAFC')

phase_labels  = [r['Phase'] for r in phase_results]
phase_scores  = [r['Predicted G3'] for r in phase_results]
phase_imps    = [r['Improvement %'] for r in phase_results]

ax3.plot(range(len(phase_labels)), phase_scores, 'o-',
         color='#2563EB', linewidth=2.5, markersize=10,
         markerfacecolor='white', markeredgewidth=2.5, zorder=4)
ax3.fill_between(range(len(phase_labels)), phase_scores,
                 min(phase_scores)-0.05, alpha=0.12, color='#2563EB')

for i, (score, imp) in enumerate(zip(phase_scores, phase_imps)):
    label = f'{score:.3f}\n({imp:+.1f}%)'
    ax3.annotate(label, (i, score), textcoords='offset points',
                 xytext=(0, 14), ha='center', fontsize=8.5,
                 fontweight='bold', color='#1E293B')

# Reference lines for other countries
for country, color in [('Finland','#10B981'),('Canada','#3B82F6')]:
    ref_val = country_profiles.loc[country,'G3_predicted']
    ax3.axhline(ref_val, color=color, linestyle='--', linewidth=1.2, alpha=0

ax3.set_xticks(range(len(phase_labels)))
ax3.set_xticklabels(phase_labels, rotation=15, ha='right', fontsize=9)
ax3.set_ylabel('Predicted G3 (Final Grade)', fontsize=10)
ax3.set_title('Phased Reform Roadmap — India Predicted Performance',
              fontsize=11, fontweight='bold', color='#1E293B')
ax3.legend(fontsize=8.5)
ax3.spines[['top','right']].set_visible(False)

# ── Chart 4: Hybrid contribution by source country ──────────────────────
ax4 = fig.add_subplot(gs[1, 1])
ax4.set_facecolor('#F8FAFC')

source_contributions = {}
for feat, source, alpha, _ in HYBRID_CONFIG:
    source_val = country_profiles.loc[source, feat]
    india_val  = india[feat]
    delta      = abs(alpha * (source_val - india_val)) * fi_norm[feat]
```

```
    source_contributions[source] = source_contributions.get(source, 0) + del

sources = list(source_contributions.keys())
contribs = list(source_contributions.values())
total    = sum(contribs)
pcts     = [c/total*100 for c in contribs]
colors   = [SOURCE_COLORS.get(s,'#64748B') for s in sources]

wedges, texts, autotexts = ax4.pie(
    pcts, labels=sources, autopct='%1.1f%%',
    colors=colors, startangle=140,
    pctdistance=0.75, labeldistance=1.12,
    wedgeprops={'edgecolor':'white','linewidth':2}
)
for t in texts: t.set_fontsize(11)
for at in autotexts: at.set_fontsize(9); at.set_fontweight('bold')
ax4.set_title('Hybrid Model — Source Country Contributions\n(weighted by fea
              fontsize=11, fontweight='bold', color='#1E293B')

# ── Chart 5: Before vs After all countries ──────────────────────────────
ax5 = fig.add_subplot(gs[2, :])
ax5.set_facecolor('#F8FAFC')

all_countries   = list(country_profiles.index)
all_preds_base  = [country_profiles.loc[c,'G3_predicted'] for c in all_count

# Add Hybrid India as a new entry
plot_countries  = all_countries + ['India (Hybrid)']
plot_preds      = all_preds_base + [hybrid_pred]

plot_df = pd.DataFrame({'Country': plot_countries, 'Predicted G3': plot_pred
plot_df = plot_df.sort_values('Predicted G3', ascending=True)

bar_colors_final = []
for c in plot_df['Country']:
    if c == 'India (Hybrid)': bar_colors_final.append('#06B6D4')
    elif c == 'India':        bar_colors_final.append('#EF4444')
    else:                     bar_colors_final.append('#94A3B8')

bars = ax5.barh(
    plot_df['Country'], plot_df['Predicted G3'],
    color=bar_colors_final, edgecolor='white', linewidth=0.5, height=0.65
)
for bar, val in zip(bars, plot_df['Predicted G3']):
    ax5.text(bar.get_width()+0.04, bar.get_y()+bar.get_height()/2,
             f'{val:.3f}', va='center', fontsize=9.5, fontweight='bold')

ax5.set_xlabel('ML Predicted G3 Score', fontsize=11)
ax5.set_title('All Countries + India Hybrid — ML Predicted Performance Compa
              fontsize=12, fontweight='bold', color='#1E293B')
ax5.spines[['top','right']].set_visible(False)
p1 = mpatches.Patch(color='#EF4444', label='India (current)')
p2 = mpatches.Patch(color='#06B6D4', label='India (hybrid model)')
p3 = mpatches.Patch(color='#94A3B8', label='Other countries')
ax5.legend(handles=[p1,p2,p3], fontsize=10, loc='lower right')
```

```
plt.savefig('module7_hybrid_model.png', dpi=150, bbox_inches='tight', faceco
plt.show()
print('✅ Chart saved as module7_hybrid_model.png')
```



✅ Chart saved as module7_hybrid_model.png

📋 Step 8 — Policy Recommendations (Report-Ready)

```
print('=' * 70)
print('  HYBRID EDUCATION REFORM MODEL — INDIA')
print('  AI-Generated Policy Recommendations')
print('=' * 70)

india_g3   = india_pred
hybrid_g3  = hybrid_pred
final_imp  = ((hybrid_g3 - india_g3) / india_g3) * 100

print(f"""
EXECUTIVE SUMMARY
-----------------
India current predicted performance : G3 = {india_g3:.4f}
Hybrid model predicted performance  : G3 = {hybrid_g3:.4f}
Expected improvement                : +{final_imp:.2f}%
""")

reforms = [
    ('REFORM 1: Teacher Quality', 'Finland/Germany',
     f"Current teacher quality index: {india['teacher_ratio_index']:.3f} | T
     ['Increase minimum teacher qualification to post-graduate level',
      'Introduce continuous professional development programs (CPD)',
      'Deploy mentorship programs pairing rural teachers with urban experts'
      'Raise teacher salaries to attract top graduates into teaching',
      'Implement Finland-style teacher autonomy in curriculum delivery']),

    ('REFORM 2: Student Well-being', 'Finland',
     f"Current wellbeing index: {india['student_wellbeing_index']:.3f} | Tar
     ['Replace high-stakes single exams with continuous formative assessment
      'Introduce school counseling and mental health support programs',
      'Regulate private coaching institutes to reduce academic pressure',
      'Add arts, sports, and creative subjects as core curriculum components
      'Adopt play-based and inquiry-based learning at primary levels']),

    ('REFORM 3: Government Education Spending', 'Canada',
     f"Current spend index: {india['govt_edu_spend_index']:.3f} | Target: {h
     ['Increase public education budget to minimum 6% of GDP',
      'Prioritize rural school infrastructure investment',
      'Fund state-of-the-art STEM laboratories in every district',
      'Provide free learning devices and internet access to all students',
      'Introduce need-based scholarship programs for higher education']),

    ('REFORM 4: Academic Standards', 'Singapore',
     f"Current math/science index: {india['math_score_index']:.3f} | Target:
     ['Restructure math and science curricula for conceptual understanding',
      'Introduce problem-based and project-based learning frameworks',
      'Develop Singapore-style structured progression in STEM subjects',
      'Create national digital learning platforms with adaptive content',
      'Strengthen international benchmarking through PISA participation']),

    ('REFORM 5: Reduce Failures & Absences', 'Finland',
```

```
        f"Current failures: {india['failures']:.3f} | Target: {hybrid['failures
        ['Implement early intervention programs for struggling students',
         'Introduce remedial classes instead of grade repetition',
         'Address root causes of absenteeism (transport, safety, poverty)',
         'Make mid-day meals and health checks universal in all schools',
         'Use attendance tracking and community engagement programs']),
]

for title, source, metric, points in reforms:
    print(f'  {title}  [Source: {source}]')
    print(f'  {metric}')
    for pt in points:
        print(f'    • {pt}')
    print()

print('PHASED IMPLEMENTATION TIMELINE')
print('-' * 50)
for i, (phase, alpha, desc) in enumerate(PHASES):
    row = phase_df[phase_df['Phase'] == phase].iloc[0]
    print(f'  {phase}: {desc}')
    print(f'     → Predicted G3: {row["Predicted G3"]:.4f} | Improvement: {r
print()
print(f'  CONCLUSION: Full implementation of the Hybrid Model')
print(f'  predicts a {final_imp:.2f}% improvement in India educational perfo
print(f'  moving India from rank 10 toward the global middle tier.')
```

```
======================================================================
  HYBRID EDUCATION REFORM MODEL — INDIA
  AI-Generated Policy Recommendations
======================================================================

EXECUTIVE SUMMARY
-----------------
India current predicted performance : G3 = 4.7350
Hybrid model predicted performance  : G3 = 6.1645
Expected improvement                : +30.19%

  REFORM 1: Teacher Quality  [Source: Finland/Germany]
  Current teacher quality index: 3.455 | Target: 6.229
    • Increase minimum teacher qualification to post-graduate level
    • Introduce continuous professional development programs (CPD)
    • Deploy mentorship programs pairing rural teachers with urban experts
    • Raise teacher salaries to attract top graduates into teaching
    • Implement Finland-style teacher autonomy in curriculum delivery

  REFORM 2: Student Well-being  [Source: Finland]
  Current wellbeing index: 4.966 | Target: 7.302
    • Replace high-stakes single exams with continuous formative assessment
    • Introduce school counseling and mental health support programs
    • Regulate private coaching institutes to reduce academic pressure
    • Add arts, sports, and creative subjects as core curriculum components
    • Adopt play-based and inquiry-based learning at primary levels

  REFORM 3: Government Education Spending  [Source: Canada]
  Current spend index: 4.088 | Target: 5.827
    • Increase public education budget to minimum 6% of GDP
    • Prioritize rural school infrastructure investment
```

```
    • Fund state-of-the-art STEM laboratories in every district
    • Provide free learning devices and internet access to all students
    • Introduce need-based scholarship programs for higher education

  REFORM 4: Academic Standards  [Source: Singapore]
  Current math/science index: 4.538 | Target: 6.021
    • Restructure math and science curricula for conceptual understanding
    • Introduce problem-based and project-based learning frameworks
    • Develop Singapore-style structured progression in STEM subjects
    • Create national digital learning platforms with adaptive content
    • Strengthen international benchmarking through PISA participation

  REFORM 5: Reduce Failures & Absences  [Source: Finland]
  Current failures: 5.959 | Target: 3.735 | Current absences: 5.541 | Target
    • Implement early intervention programs for struggling students
    • Introduce remedial classes instead of grade repetition
    • Address root causes of absenteeism (transport, safety, poverty)
    • Make mid-day meals and health checks universal in all schools
    • Use attendance tracking and community engagement programs

PHASED IMPLEMENTATION TIMELINE
--------------------------------------------------
  Phase 1 (0-2 yrs): Policy foundation: teacher recruitment & wellbeing prog
     → Predicted G3: 4.6381 | Improvement: -2.05%
  Phase 2 (2-4 yrs): Infrastructure: school quality & spending increase
```